The empirical recurrence for OEIS A316279, recovered from its tail

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Abstract

OEIS A316279 records "Empirical recurrence of order 95" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 160$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{95} = -64 \neq 0$, so no further tail satisfies anything shorter; any order-95 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316279 is "Number of nX5 0..1 arrays with every element unequal to 1, 2, 3, 4, 5 or 8 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$3, 383, 5321, 102072, 1820908, 32932624, 594230641, 10728510594, \dots$$

The entry states:

> Empirical recurrence of order 95 (see link above)

As of the "Last modified" line on the live entry (Jun 28 2018 15:24:02, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 160$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 160$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 95: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 95.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 320$ exact terms from $n = 3$ on, it returns order 95 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{95} c_i\, a(n-i),$$

c_1	10	c_{25}	−537492024108	c_{49}	−305083648864269	c_{73}	182565238950
c_2	212	c_{26}	−1064803682206	c_{50}	219145104178498	c_{74}	−113135493976
c_3	−220	c_{27}	−4036781507205	c_{51}	92229982980077	c_{75}	−202890823890
c_4	−15684	c_{28}	−29291869299	c_{52}	1100748959015783	c_{76}	−151625545910
c_5	−60195	c_{29}	−4112100923191	c_{53}	586213575598587	c_{77}	−131014352101
c_6	270710	c_{30}	8971521407422	c_{54}	598208303024303	c_{78}	−95093290950
c_7	2573358	c_{31}	6722213414028	c_{55}	359147784915892	c_{79}	−59537722845
c_8	3586912	c_{32}	7887953891241	c_{56}	70415542816044	c_{80}	−33711907692
c_9	−28545769	c_{33}	27815210648421	c_{57}	390708896249190	c_{81}	−15274808387
c_{10}	−125101328	c_{34}	−21748620800249	c_{58}	−58398952869878	c_{82}	−6329725723
c_{11}	−70964334	c_{35}	38558253665128	c_{59}	−47346345568755	c_{83}	−2034259973
c_{12}	790418798	c_{36}	−63978292569432	c_{60}	−80195586514117	c_{84}	−779349494
c_{13}	2550590894	c_{37}	−3712843187650	c_{61}	−16873697654535	c_{85}	−224109857
c_{14}	1943875012	c_{38}	−39109303893093	c_{62}	136747063819728	c_{86}	−10010476
c_{15}	−7520818026	c_{39}	−60814080781438	c_{63}	110382853115409	c_{87}	9244006
c_{16}	−24823585846	c_{40}	39590696824176	c_{64}	70217375857620	c_{88}	15695409
c_{17}	−31436487199	c_{41}	14117160034966	c_{65}	23017779149799	c_{89}	3855987
c_{18}	5704385625	c_{42}	151959069628335	c_{66}	−3697152918992	c_{90}	953547
c_{19}	87003142697	c_{43}	−49374303497558	c_{67}	−11788237832892	c_{91}	−266103
c_{20}	138508930782	c_{44}	−30354415126226	c_{68}	−13744872908560	c_{92}	22424
c_{21}	167722168774	c_{45}	60972786666250	c_{69}	−13057334634380	c_{93}	16052
c_{22}	−2096816666	c_{46}	−290836530377357	c_{70}	−6836531307735	c_{94}	3712
c_{23}	144566329342	c_{47}	−142214147803560	c_{71}	−2315781740682	c_{95}	−64
c_{24}	−151226090516	c_{48}	−327670580575438	c_{72}	−540380282469		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A316279 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{95} - c_1 x^{95-1} - \cdots - c_{95}$. Its constant term is $-c_{95} = 64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 95 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 95, so $\deg q = 95$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 95, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 95, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -64.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-95 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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